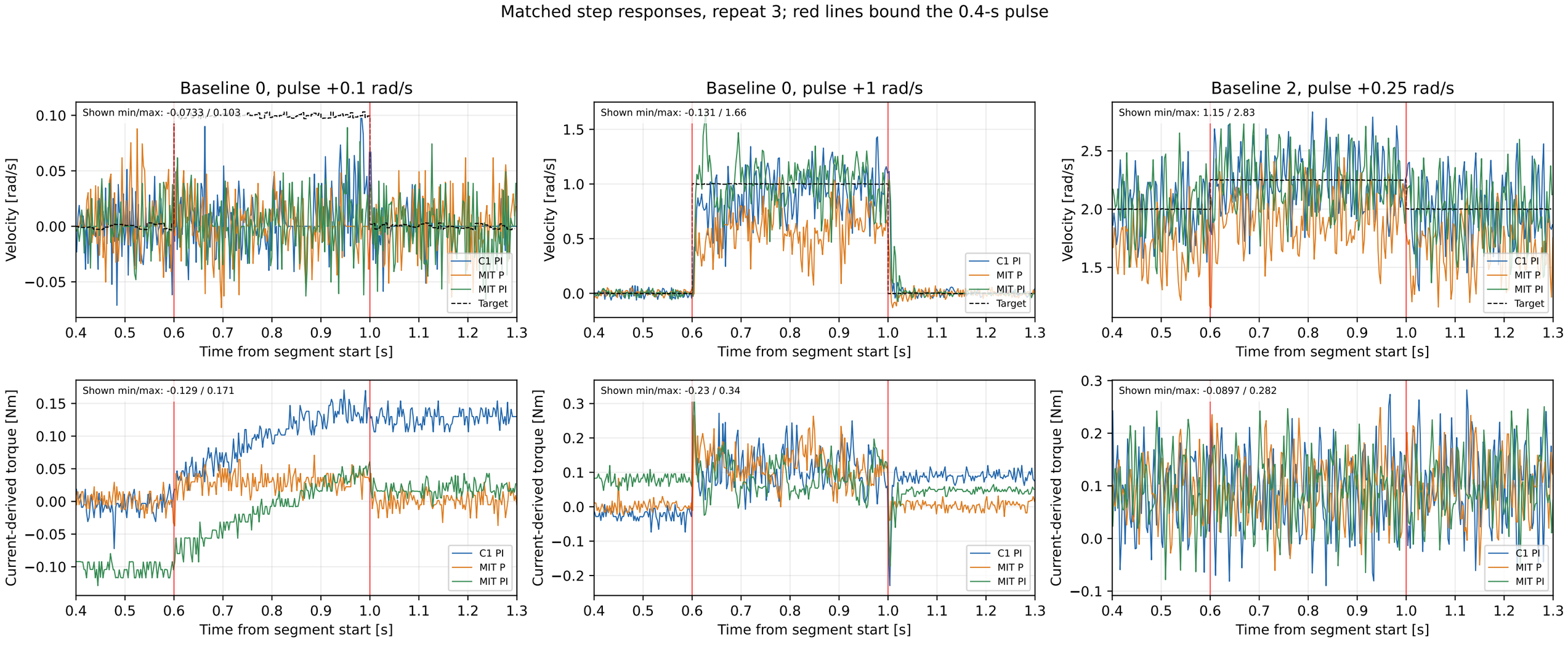

Matched step responses, repeat 3; red lines bound the 0.4-s pulse
Baseline 0, pulse +0.1 rad/s
Shown min/max: -0.0733 / 0.103
Baseline 0, pulse +1 rad/s
Shown min/max: -0.131 / 1.66
Baseline 2, pulse +0.25 rad/s
Shown min/max: 1.15 / 2.83
Velocity [rad/s]
Current-derived torque [Nm]
Shown min/max: -0.129 / 0.171
Shown min/max: -0.23 / 0.34
Shown min/max: -0.0897 / 0.282
Time from segment start [s]
C1 PI
MIT P
MIT PI
Target
Shown 0.4–1.3 s to resolve the pulse after the initial baseline transient. Integral resets once per segment. Native A2/A1; no smoothing/time shift.